Simulation at scale

NAMES _____ CLASS _____

STEP 1 — SET UP

My chance device is:

The outcome I am following is: Its probability is:

Step 2 · Run it at every scale

Predict and lock in before each run. Then fill in this row.

TRIALS	OBSERVED	EXPECTED	DIFFERENCE observed – expected	RELATIVE FREQUENCY observed ÷ trials
20				
200				
2,000				

THE NEW WORD TODAY — RELATIVE FREQUENCY

Relative frequency = observed ÷ trials.
It turns your count into a number you can compare straight to the probability.
Example: 11 reds out of 20 trials → 11 ÷ 20 = 0.55

BEFORE YOU TURN OVER

Look down the DIFFERENCE column. Is it getting bigger or smaller?
Now look down the RELATIVE FREQUENCY column. Is it getting closer to the probability?
Those two answers are not the same. That is not a mistake — it is the point of today.

WORKING SPACE

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Part B · Show your thinking

SF

1. Out of 50 spins, a spinner landed on red 21 times.

What is the relative frequency? Give it as a decimal.

SF

2. The probability was 0.5.

Is 0.42 close to 0.5? Write yes or no, and say why.

CF

3. Which run gives you better evidence: 20 trials or 2000 trials?

Explain your answer.

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If the devices are not working

These are real results from a fair coin. You can do all of today's thinking from them.

TRIALS	OBSERVED	EXPECTED	DIFFERENCE observed – expected	RELATIVE FREQUENCY observed ÷ trials
20	11			
200	96			
2,000	1,013			
20,000	10,024			

The probability of heads is 0.5. Fill in the other three columns yourself.

Exit ticket

Hand this in at the end of the lesson.

1. A simulator ran a fair coin 100 times and got 62 heads.

What is the relative frequency of heads? Give it as a decimal.

2. Is 62 heads close to what you would expect? Explain.

3. Jo ran the same coin 10 times and got 6 heads.

Whose result is better evidence that the coin is fair? Explain.

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